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P5 — Radioactive Materials

OCR GCSE Physics B (J259) — 21st Century Science · Paper 2 (Depth in Physics)

P5.1 The Atomic Model — A Historical Story

Ideas about the atom have changed over time as experimental evidence accumulated. You should know the sequence of models:

Model	Scientist(s)	Key idea
Tiny indivisible particles ("atomos")	Democritus (≈ 400 BC)	Matter made of indivisible units
Sphere of positive matter with negative electrons dotted through it	J.J. Thomson, 1897 — **plum pudding model**	Discovery of the electron
Small, dense, positive nucleus with electrons around it	Rutherford, 1911 — **gold foil / α-scattering experiment**	Most of the atom is empty space; nearly all its mass is in a tiny central nucleus
Electrons orbit in fixed shells (energy levels)	Bohr, 1913	Explains line spectra
Nucleus contains protons and neutrons	Chadwick, 1932	Discovery of the neutron

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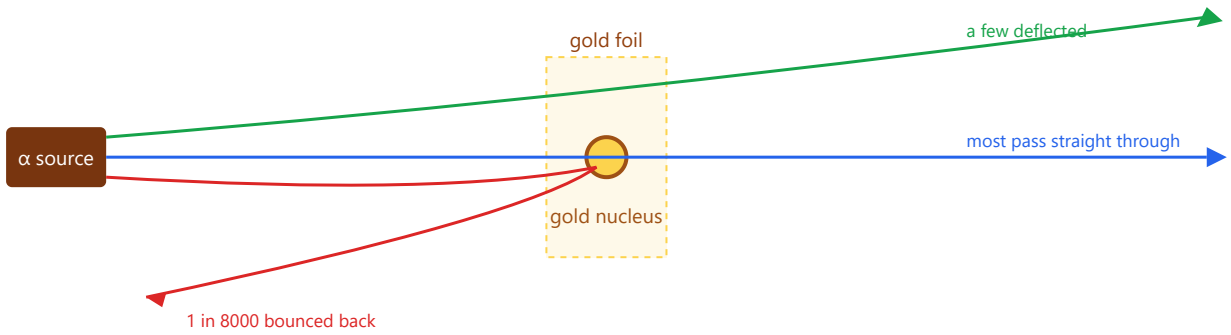
Rutherford's α-particle Scattering Experiment

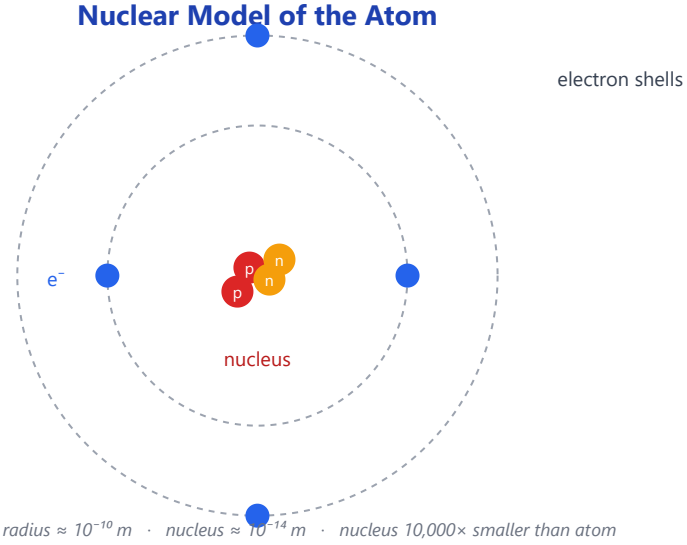


Exam Tip: Rutherford's **conclusions** from the results:

- Most α pass straight through → atoms are mostly empty space.
- A few deflected at large angles → small, dense, *positively charged* nucleus.
- 1 in 8000 bounced back → nucleus is very small but contains nearly all the mass.

P5.2 Structure of the Atom

Nuclear Model of the Atom



Particle	Relative mass	Relative charge	Location
Proton (p)	1	+1	Nucleus

Neutron (n)	1	0	Nucleus
Electron (e^-)	1/1836 (negligible)	−1	Shells / orbitals

Nuclear Notation

An atom is written as $^{A}_{Z}X$ where:

- **Z** = proton (atomic) number — number of protons; defines the element.
- **A** = mass (nucleon) number — total protons + neutrons.
- Number of neutrons = A − Z.

Isotopes are atoms of the same element with the same Z but different A (different number of neutrons). Example: hydrogen has three isotopes — ^{1}H, ^{2}H (deuterium), ^{3}H (tritium).

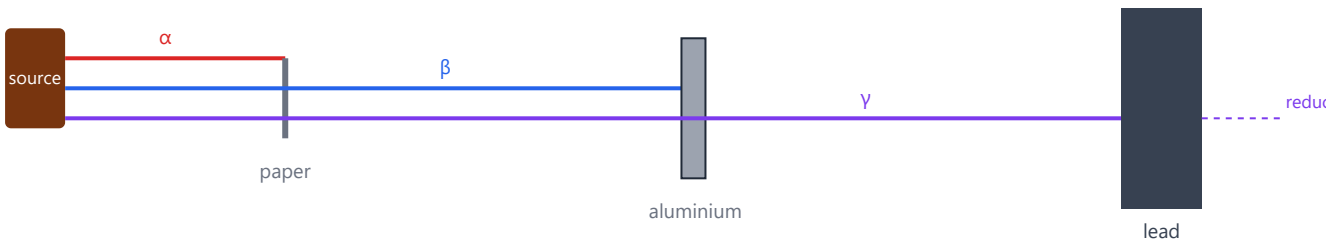
P5.3 Types of Nuclear Radiation

> Some nuclei are **unstable** — they decay spontaneously, emitting radiation. The three main types you need are **alpha (α), beta (β) and gamma (γ)**. Positrons (β^+) and free neutrons are also mentioned at GCSE.

Radiation	What it is	Charge	Mass	Stopped by	Ionising
Alpha (α)	Helium nucleus: $^{4}_{2}He$ — 2 p + 2 n	+2	4 u	A few cm of air; paper; skin	Very strongly ionising
Beta-minus (β^-)	Fast electron from the nucleus (a neutron → proton + e^-)	−1	≈ 0	A few mm of aluminium	Moderately ionising
Gamma (γ)	High-frequency EM wave	0	0	Several cm of lead / metres of concrete (reduced, never fully)	Weakly ionising

Positron (β^+)	Antiparticle of the electron (proton → neutron + β^+)	+1	≈ 0	Annihilates with an electron almost immediately	—
Neutron	Free neutron (some decays)	0	1	Water / hydrogen-rich materials	Via induced radioactivity

Penetration of α, β and γ



P5.4 Nuclear Equations

α decay: $^{A}_{Z}X \rightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}He$

(mass falls by 4, atomic number falls by 2)

β^- decay: $^{A}_{Z}X \rightarrow {}^{A}_{Z+1}Y + {}^{0}_{-1}e$

(mass unchanged, atomic number rises by 1 — a neutron becomes a proton)

γ emission: no change in A or Z — the nucleus loses energy as a γ photon.

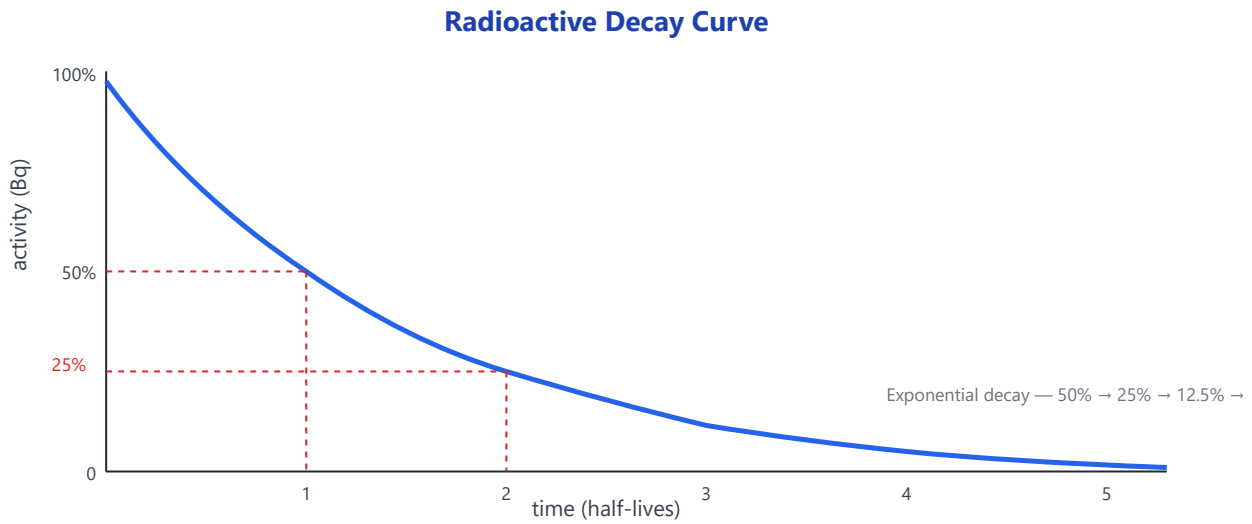
Worked Example: Uranium-238 decays by alpha emission. Write the nuclear equation.
$^{238}_{92}U \rightarrow {}^{234}_{90}Th + {}^{4}_{2}He$ (the daughter is thorium-234).

Worked Example: Carbon-14 decays by β^- emission. Write the nuclear equation.

$^{14}_{6}C \rightarrow {}^{14}_{7}N + {}^{0}_{-1}e$

P5.5 Activity and Half-Life

Activity (A) = number of decays per second. Unit: **becquerel (Bq)**, 1 Bq = 1 decay/s.
Half-life ($t_{1/2}$) = the average time for the number of unstable nuclei in a sample — or equivalently the activity — to halve.

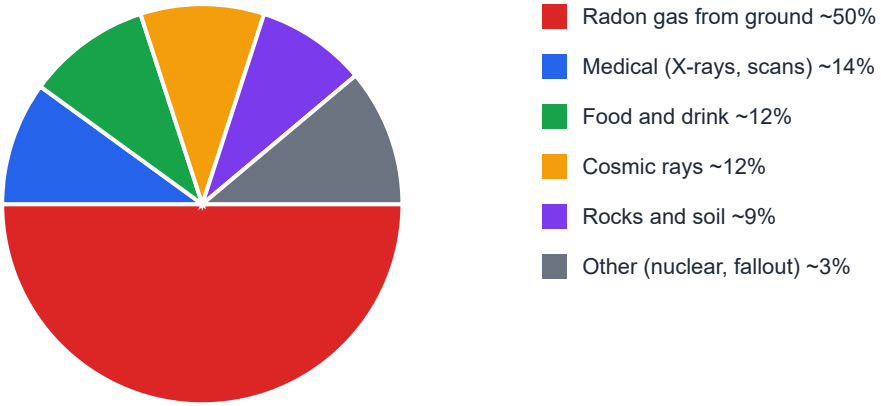


Worked Example: A sample of iodine-131 has an initial activity of 8000 Bq. Its half-life is 8 days. Find the activity after 24 days.
24 ÷ 8 = 3 half-lives. Activity = 8000 × $(½)^3$ = 8000 / 8 = **1000 Bq**.

P5.6 Background Radiation

There is always a low level of ionising radiation around us. Typical UK sources (approximate contributions):

Sources of UK Background Radiation



Background radiation must be **subtracted** from measured count rates when taking readings from a source, otherwise results are too high.

P5.7 Uses of Radiation in Medicine & Industry

Use	Radiation type	Why that type
Medical tracer (e.g. technetium-99m)	γ + short half-life (6 h)	Leaves body quickly; penetrates tissue to the detector
Radiotherapy for cancer	γ or high-energy X-ray (external); β (internal)	Kills cancer cells; directed from many angles to minimise healthy tissue dose
Sterilising surgical equipment / food	γ	Penetrates packaging; kills bacteria
Smoke detectors	α (americium-241)	Short range — safe in the home; smoke absorbs α, breaking a small ionisation current
Thickness gauge for paper / foil	β	Partially absorbed — change in transmitted count indicates thickness

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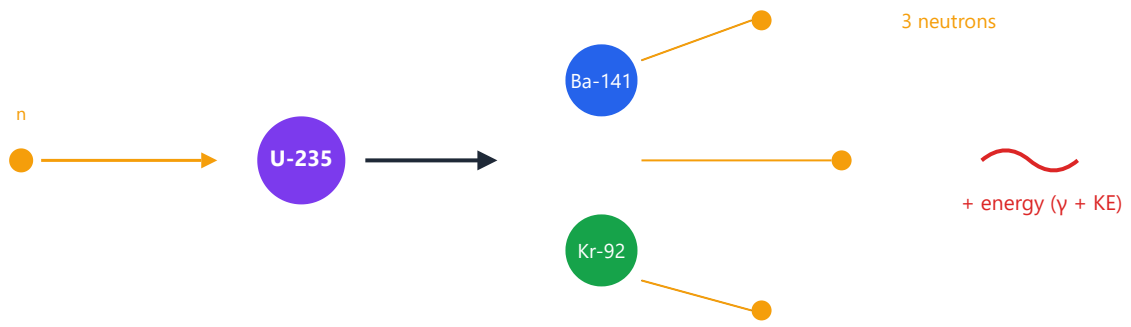
Dating rocks / archaeology	U-238, C-14	Long, known half-lives

P5.8 Nuclear Fission and Fusion

Fission: a large unstable nucleus (e.g. U-235) splits into two smaller, more stable nuclei, releasing a few neutrons and energy. The neutrons can trigger further fissions — a **chain reaction**.

Nuclear Fission of U-235



Fusion: two small nuclei (e.g. hydrogen isotopes) collide at very high speed and join to make a larger nucleus (helium), releasing energy. This is how the Sun and all stars shine.

Fusion releases *much more energy per kg* than fission but requires extremely high temperatures and pressures to overcome electrostatic repulsion between the nuclei. Sustainable Earth-based fusion power is still being developed (ITER, tokamak designs).

P5.9 Dangers of Ionising Radiation

- **Irradiation**: being exposed to radiation *from outside*. The object does not become radioactive.
- **Contamination**: radioactive material gets *onto or into* the body (skin, food, dust). Continues to irradiate from inside — much harder to stop.

Risk depends on the activity, the type of radiation, and the duration and location of exposure. Inside the body, **alpha** is the most dangerous because it is very strongly ionising; outside the body, **gamma** is the most dangerous because it penetrates the most.

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